heads as high as 0.8, but as you keep flipping, it will tend toward 0.5 exactly, in the infinite limit.

Hence, probabilities are just frequencies of events. In this case, there are two possible outcomes, heads or tails, and each outcome's probability is its frequency after an infinite number of trials (coin flips). This is, of course, why probabilities are values between 0 (impossible) and 1 (certain), and the probabilities for all possible outcomes must sum to 1.

This is a simple and straightforward approach to probability, but it has significant limitations. In the frequentist setting, it is difficult or perhaps impossible to make sense of a question like "what is the probability that Jane Doe will be elected to city council?" since it is impossible in practice and theory for such an election to happen an infinite number of times. Frequentist probability doesn't make much sense for these kinds of one-off events. We need a more powerful framework to handle these situations, and that is what Bayesian probability gives us.

In the Bayesian framework, probabilities represent degrees of belief about various possible outcomes. You can certainly have a belief about something that can only happen once, like an election, and your belief about what is likely to happen can vary depending on how much information you have about a particular situation, and new information will cause you to update your beliefs (see table 7.1).

Table 7.1. Frequentist versus Bayesian probabilities

Frequentist	Bayesian
Probabilities are frequencies of individual outcomes	Probabilities are degrees of belief
Computes the probability of the data given a model	Computes the probability of a model given the data
Uses hypothesis testing	Uses parameter estimation or model comparison
Is computationally easy	Is (usually) computationally difficult

The basic mathematical framework for probability consists of a *sample space*, Ω, which is the set of all possible outcomes for a particular question.